

Module 11: Genomics and Biotechnology — Study Questions

1. What is a restriction enzyme? How do "sticky ends" facilitate ligation?
2. What is a vector (e.g., plasmid)? How is it used to introduce foreign DNA into bacteria?
3. Describe the three steps of PCR: Denaturation, Annealing, Extension.
4. Why is PCR called "molecular photocopying"? Why is it useful for forensics?
5. Explain how gel electrophoresis separates DNA fragments.
6. Do smaller fragments migrate faster or slower? Why?
7. Define transgenic organism (GMO).
8. Outline the steps to produce insulin-secreting bacteria.
9. In a paternity test, a child has bands at positions A and B. The mother has band A. Must the biological father have band B? Explain.
10. Differentiate ex vivo (cells treated outside the body) and in vivo (treatment inside the body) gene therapy.
11. What are the major challenges of gene therapy? (Delivery, immune response, targeting)
12. We have sequenced the human genome. Why doesn't knowing the sequence immediately reveal gene function?
13. What is functional genomics?
14. What is bioinformatics? How do computational tools help analyze biological data?
15. What are STRs (Short Tandem Repeats)? How are they used in genetic fingerprinting?

16. Compare the production of recombinant DNA with traditional selective breeding. What advantages does genetic engineering offer?
17. Golden Rice is engineered to produce beta-carotene (Vitamin A precursor). Explain why this is considered a humanitarian application of biotechnology.
18. What is CRISPR-Cas9? How does it differ from traditional restriction enzyme methods of gene editing?
19. Discuss the ethical implications of human germline editing. What is the difference between somatic gene therapy and germline modification?
20. The genetic code is nearly universal. How does this support common ancestry? How does it enable biotechnology (e.g., producing human insulin in bacteria)?